

1.

```
create database insuranceDB;
```

```
Commands completed successfully.
```

2.

```
--customers table
```

```
create table customers (  
    customerId int primary key,  
    firstName varchar(10) null,  
    lastName varchar(50) not null,  
    dateOfBirth date not null,  
    phone varchar(12) not null,  
    email varchar(50) not null  
);
```

```
--policies table
```

```
create table policies(  
    policyId int primary key,  
    policyName varchar(20),  
    policyType varchar(20),  
    premiumAmount decimal(10,2),  
    durationYears int  
)
```

```
--claims table
```

```
create table claims(  
    claimId int primary key,  
    assignmentId int,  
    claimDate date,  
    claimAmount decimal(10,2),  
    claimStatus varchar(5),  
    foreign key(assignmentId) references policyAssignments(assignmentId)  
)
```

```
--agents table
```

```
create table agents(  
    agentId int primary key,  
    agentName varchar(50),  
    phone varchar(12),  
    city varchar(50)  
)
```

```
--policy assignments
```

```
create table policyAssignments(  
    assignmentId int primary key,  
    customerId int,  
    policyId int,  
    agentId int,
```

```

startDate date,
endDate date
foreign key (customerId) references customers(customerId),
foreign key (policyId) references policies(policyId),
foreign key (agentId) references agents(agentId)
)

```

3.

--insert customers

```

insert into customers values(1,'Amit','Sharma','1995-10-10','9989727567','amit@mail.com'),
(2,'preethi','varma','1998-09-08','8989898989','preethi@mail.com')

```

--insert agents

```

insert into agents values(101,'Santhosh','9989898988','Hyd')

```

--insert policies

```

insert into policies values(120,'Health Secure','Health',12000.00,1),
(121,'Life Shield','Life',25000.00,10),
(122,'Motor Protect','Motor',8000.00,1)

```

--insert policy assignments

```

insert into policyAssignments values(202,1,120,101,'2020-10-10','2020-12-10')

```

--insert claims

```

insert into claims values(1000,202,'2020-10-10',10000,'Yes')

```

4.

1.select * from customers;

	customerId	firstName	lastName	dateOfBirth	phone	email
1	1	Amit	Sharma	1995-10-10	9989727567	amit@mail.com
2	2	preethi	varma	1998-09-08	8989898989	preethi@mail.com
3	3	Vishwa	Nair	2014-10-10	8976598765	vishwa@mail.com

2. select customerId,policyId,startDate,endDate from policyAssignments;

	customerId	policyId	startDate	endDate
1	1	120	2020-10-10	2020-12-10
2	2	120	2021-09-10	2021-10-12
3	1	121	2021-09-10	2021-12-11

3. select * from policies where policyType='Health';

	policyId	policyName	policyType	premiumAmount	durationYears
1	120	Health Secure	Health	12000.00	1

4. select * from policies where premiumAmount>10000 and durationYears=1;

	policyId	policyName	policyType	premiumAmount	durationYears
1	120	Health Secure	Health	12000.00	1

5. select distinct city from agents;

	city
1	Delhi
2	Hyderabad
3	Kakinada

6. select * from policies where policyType='Health' or policyType='Life' or policyType='Motor';

	policyId	policyName	policyType	premiumAmount	durationYears
1	120	Health Secure	Health	12000.00	1
2	121	Life Shield	Life	25000.00	10
3	122	Motor Protect	Motor	8000.00	1

7. select * from policies where policyType in ('Health','Life','Motor');

	policyId	policyName	policyType	premiumAmount	durationYears
1	120	Health Secure	Health	12000.00	1
2	121	Life Shield	Life	25000.00	10
3	122	Motor Protect	Motor	8000.00	1

8. select * from customers where dateOfBirth>='2001-1-1' and dateOfBirth<='2020-12-31';

	customerId	firstName	lastName	dateOfBirth	phone	email
1	3	Vishwa	Nair	2014-10-10	8976598765	vishwa@mail.com

9. select * from customers where dateOfBirth between '2001-1-1' and '2020-12-31';

	customerId	firstName	lastName	dateOfBirth	phone	email
1	3	Vishwa	Nair	2014-10-10	8976598765	vishwa@mail.com

10. select * from claims where claimStatus='Rejected';

	claimId	assignmentId	claimDate	claimAmount	claimStatus
1	1000	202	2020-10-11	10000.00	Rejected

11. select * from agents where city like '_a%';

	agentId	agentName	phone	city
1	104	Hithesh	8338334564	Kakinada

12. select max(claimAmount) as maxClaimAmount,min(claimAmount) as minClaimAmount from claims;

	maxClaimAmount	minClaimAmount
1	20000.00	10000.00

13. select max(claimDate) as latestDate from claims;

	latestDate
1	2020-12-11

14. update policies set premiumAmount=premiumAmount*1.1 where policyType='Health';
select * from policies where policyType='Health';

	policyId	policyName	policyType	premiumAmount	durationYears
1	120	Health Secure	Health	13200.00	1

15. select * from policyAssignments;

	assignmentId	customerId	policyId	agentId	startDate	endDate
1	201	2	120	101	2020-10-10	2021-11-11
2	202	1	120	101	2020-10-10	2026-12-10
3	205	5	122	103	2025-12-12	2026-12-12
4	206	5	121	102	2024-11-10	2025-12-31
5	207	5	120	102	2025-11-12	2026-11-17

delete from policyAssignments where endDate<=GETDATE();

select * from policyAssignments;

	assignmentId	customerId	policyId	agentId	startDate	endDate
1	202	1	120	101	2020-10-10	2026-12-10
2	205	5	122	103	2025-12-12	2026-12-12
3	206	5	121	102	2024-11-10	2025-12-31
4	207	5	120	102	2025-11-12	2026-11-17

(1 row affected)

Completion time: 2025-12-29T23:29:30.2829232+05:30

16. select count(*) as noOfClaimsRejected from claims where claimStatus='Rejected';

	noOfClaimsRejected
1	1

17. select policyId, policyName, premiumAmount, premiumAmount * 0.06 as localTaxes,
premiumAmount + (premiumAmount * 0.06) as premiumAmountWithTax, (premiumAmount +
(premiumAmount * 0.06)) / 12 as monthlyPremiumAmount from policies;

	policyId	policyName	premiumAmount	localTaxes	premiumAmountWithTax	monthlyPremiumAmount
1	120	Health Secure	14520.00	871.2000	15391.2000	1282.6000000
2	121	Life Shield	25000.00	1500.0000	26500.0000	2208.3333333
3	122	Motor Protect	8000.00	480.0000	8480.0000	706.6666666

18. alter table customers add address varchar(50);

alter table customers add city varchar(20);

Commands completed successfully.

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	customerId	firstName	lastName	dateOfBirth	phone	email	address	city
1	1	Amit	Sharma	1995-10-10	9989727567	amit@mail.com	NULL	NULL
2	2	preethi	varma	1998-09-08	8989898989	preethi@mail.com	NULL	NULL
3	3	Vishwa	Nair	2014-10-10	8976598765	vishwa@mail.com	NULL	NULL

19.

alter table agents add DevOfId int;

select * from agents;

	agentId	agentName	phone	city	DevOfId
1	101	Santhosh	9989898988	Hyderabad	NULL
2	102	Ravi	9765898756	Delhi	NULL
3	103	Girish	8978789789	Hyderabad	NULL
4	104	Hithesh	8338334564	Kakinada	NULL

20. alter table agents add constraint rfk_devofid_agentid foreign key (DevOfId) references agents(agentId);

Commands completed successfully.

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	agentId	agentName	phone	city	DevOfId
1	101	Santhosh	9989898988	Hyderabad	NULL
2	102	Ravi	9765898756	Delhi	NULL
3	103	Girish	8978789789	Hyderabad	NULL
4	104	Hithesh	8338334564	Kakinada	NULL

5.

1. select policyName as policiesForCustomer5 from policies join policyAssignments on policies.policyId=policyAssignments.policyId where customerId=5;

	policiesForCustomer5
1	Motor Protect
2	Life Shield

2. select concat(firstName,lastName) as name, policyName from customers join policyAssignments on customers.customerId=policyAssignments.customerId join policies on policyAssignments.policyId=policies.policyId;

	name	policyName
1	AmitSharma	Health Secure
2	JulienGou	Motor Protect
3	JulienGou	Life Shield

3. select concat(firstName,lastName) as name, claimId,claimAmount,claimStatus from claims join policyAssignments on claims.assignmentId=policyAssignments.assignmentId join customers on policyAssignments.customerId=customers.customerId;

	name	claimId	claimAmount	claimStatus
1	AmitSharma	1000	10000.00	Rejected
2	JulienGou	1001	20000.00	Accepted

4. select firstName, policyName, agentName, startDate, endDate from customers join policyAssignments on customers.customerId=policyAssignments.customerId join agents on policyAssignments.agentId=agents.agentId join policies on policies.policyId=policyAssignments.policyId;

	firstName	policyName	agentName	startDate	endDate
1	Amit	Health Secure	Santhosh	2020-10-10	2026-12-10
2	Julien	Motor Protect	Girish	2025-12-12	2026-12-12
3	Julien	Life Shield	Ravi	2024-11-10	2025-12-31

5. select firstName, policyName, claimAmount, claimStatus, claimDate from claims join policyAssignments pa on pa.assignmentId=claims.assignmentId join customers on pa.customerId=customers.customerId join policies on policies.policyId=pa.policyId;

	firstName	policyName	claimAmount	claimStatus	claimDate
1	Amit	Health Secure	10000.00	Rejected	2020-10-11
2	Julien	Motor Protect	20000.00	Accepted	2023-11-12

6. select firstName, policyName from customers left join policyAssignments pa on pa.customerId=customers.customerId left join policies p on p.policyId=pa.policyId;

	firstName	policyName
1	Amit	Health Secure
2	preethi	NULL
3	Vishwa	NULL
4	Hansy	NULL
5	Julien	Motor Protect
6	Julien	Life Shield

7. select firstName from customers left join policyAssignments as pa on pa.customerId=customers.customerId left join claims c on c.assignmentId=pa.assignmentId where pa.assignmentId is null;

	firstName
1	preethi
2	Vishwa
3	Hansy

8. select c.customerId,concat(c.firstName,c.lastName) as fullName,sum(claimAmount) from customers c join policyAssignments pa on c.customerId=pa.customerId join claims cl on cl.assignmentId=pa.assignmentId group by c.customerId,c.firstName,c.lastName;

	customerId	fullName	totalClaimAmount
1	1	AmitSharma	10000.00
2	5	JulienGou	75000.00

9. select c.customerId,concat(c.firstName,c.lastName) as fullName,sum(claimAmount) as totalClaimAmount from customers c join policyAssignments pa on c.customerId=pa.customerId join claims cl on cl.assignmentId=pa.assignmentId group by c.customerId,c.firstName,c.lastName having sum(claimAmount)>50000;

	customerId	fullName	totalClaimAmount
1	5	JulienGou	75000.00

10. select pa.agentId,agentName,count(*) as policyCount from agents join policyAssignments pa on pa.agentId=agents.agentId group by pa.agentId,agentName;

	agentId	agentName	policyCount
1	101	Santhosh	1
2	102	Ravi	2
3	103	Girish	1